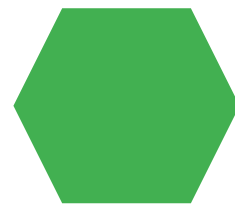




Project Title - Seasonal Agriculture Performance Analysis

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AICTE STU ID : STU6a650085cd22b1785004165



PROBLEM STATEMENT

Agricultural activities are influenced by seasonal variations in environmental conditions, farming practices, resource availability and market conditions. As a result, agricultural performance may differ from one season to another. However, raw agricultural data does not clearly explain how agricultural performance changes across seasons or what patterns can be observed in different seasonal conditions. The problem is to analyze the given agricultural dataset and investigate seasonal differences in agricultural performance by identifying meaningful patterns, trends, relationships and variations within the available data.



Project Description

- **Project Overview:** An in-depth Data Analysis of 4,000 farm records to evaluate agricultural performance across the Kharif, Rabi, and Zaid seasons.
- **Methodology:** Utilized Python (Pandas, Seaborn) to clean raw data, handle missing values, and apply statistical and visualization techniques.
- **Key Focus Areas:** Investigated how seasonal variations in environmental conditions (like rainfall and temperature) and resource usage (like water and fertilizer) impact crop yield and profitability.
- **Ultimate Goal:** To identify meaningful patterns and provide evidence-based, data-driven recommendations to support better seasonal agricultural planning.

WHO ARE THE END USERS?



Farmers & Farm Managers: To make data-driven decisions on which crops to plant in specific seasons and how to efficiently manage expensive resources like water and fertilizer.



Agricultural Planners & Policymakers: To understand environmental trends and support evidence-based agricultural planning for different regions.



Agri-Businesses & Investors: To track economic outcomes, understand market conditions, and evaluate the overall profitability of crops across the Kharif, Rabi, and Zaid seasons.

Technology Used

- **Environment:** [Google Colab](#) – Used as the primary workspace to document and run the complete analysis.
- **Programming Language:** [Python 3](#) – The core coding language used to process the data and perform calculations.
- **Data Manipulation:** [Pandas & NumPy](#) – Python libraries used to clean the missing data, group the seasonal records, and calculate the statistics.
- **Data Visualization:** [Matplotlib & Seaborn](#) – Graphing tools used to generate the colorful charts, heatmaps, and boxplots for this presentation.



RESULTS



```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt

df=pd.read_csv("/seasonal_agriculture_performance_dataset.csv")
```

```
print(f"Dataset Shape {df.shape}")
```

Dataset Shape (4000, 28)

```
print("\n--- Missing Values Check ---")
print(df.isnull().sum()[df.isnull().sum() > 0])
```

```
--- Missing Values Check ---
Rainfall_mm      48
Soil_Moisture_pct 40
Yield_Tonnes_Ha   32
dtype: int64
```

```
display(df.head())
```

	Farm_ID	State	District	Crop	Season	Farm_Area_Hectares	Rainfall_mm
0	SF10001	Andhra Pradesh	Rajkot	Wheat	Kharif	0.53	486.4
1	SF10002	Maharashtra	Nalgonda	Maize	Kharif	6.53	855.4
2	SF10003	Telangana	Warangal	Pulses	Rabi	4.86	455.6
3	SF10004	Telangana	Indore	Rice	Kharif	4.50	753.2
4	SF10005	Karnataka	Ludhiana	Maize	Zaid	4.21	101.6

5 rows × 29 columns

```
df['Rainfall_mm']=df['Rainfall_mm'].fillna(df['Rainfall_mm'].mean())
df['Soil_Moisture_pct']=df['Soil_Moisture_pct'].fillna(df['Soil_Moisture_pct'].mean())
df['Yield_Tonnes_Ha'] = df['Yield_Tonnes_Ha'].fillna(df['Yield_Tonnes_Ha'].mean())
```

```
remaining_miss=df.isnull().sum().sum()
print(f"Total missing values remaining: {remaining_miss}")
```

Total missing values remaining: 0

```
sns.set_theme(style="whitegrid", palette="muted")
print("Data is clean and ready for analysis!")
```



RESULTS

```
df['Rainfall_mm']=df['Rainfall_mm'].fillna(df['Rainfall_mm'].mean())
df['Soil_Moisture_pct']=df['Soil_Moisture_pct'].fillna(df['Soil_Moisture_pct'].mean())
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Data is clean and ready for analysis!

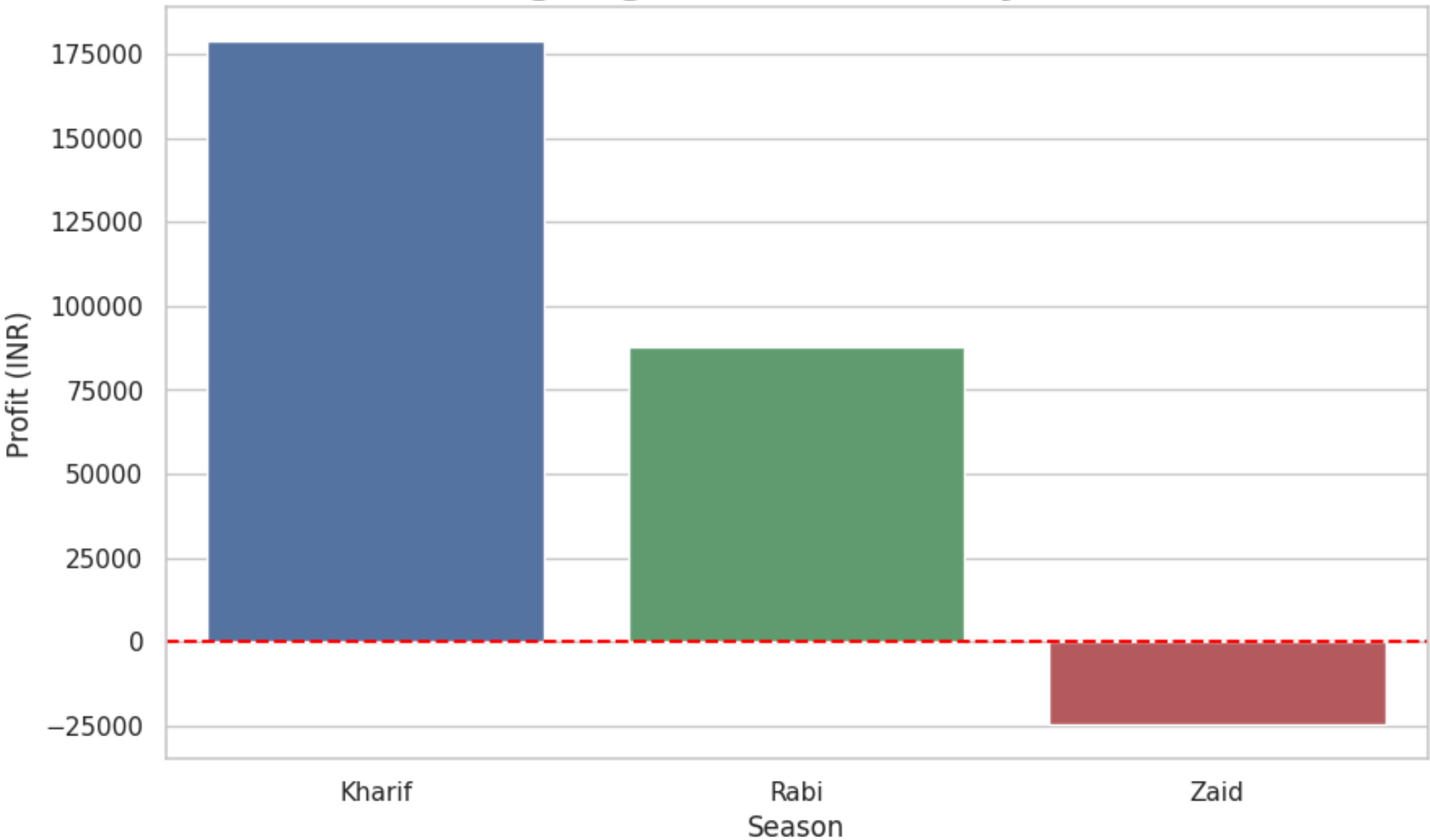
```
eco_metrics = ['Total_Cost_INR', 'Revenue_INR', 'Profit_INR']
seasonal_eco = df.groupby('Season')[eco_metrics].mean().reset_index()
print("--- Average Economic Outcomes by Season ---")
display(seasonal_eco.round(2))
```

--- Average Economic Outcomes by Season ---

	Season	Total_Cost_INR	Revenue_INR	Profit_INR
0	Kharif	531804.41	710719.06	178914.65
1	Rabi	513836.58	601526.05	87689.47
2	Zaid	543976.73	519171.90	-24804.82

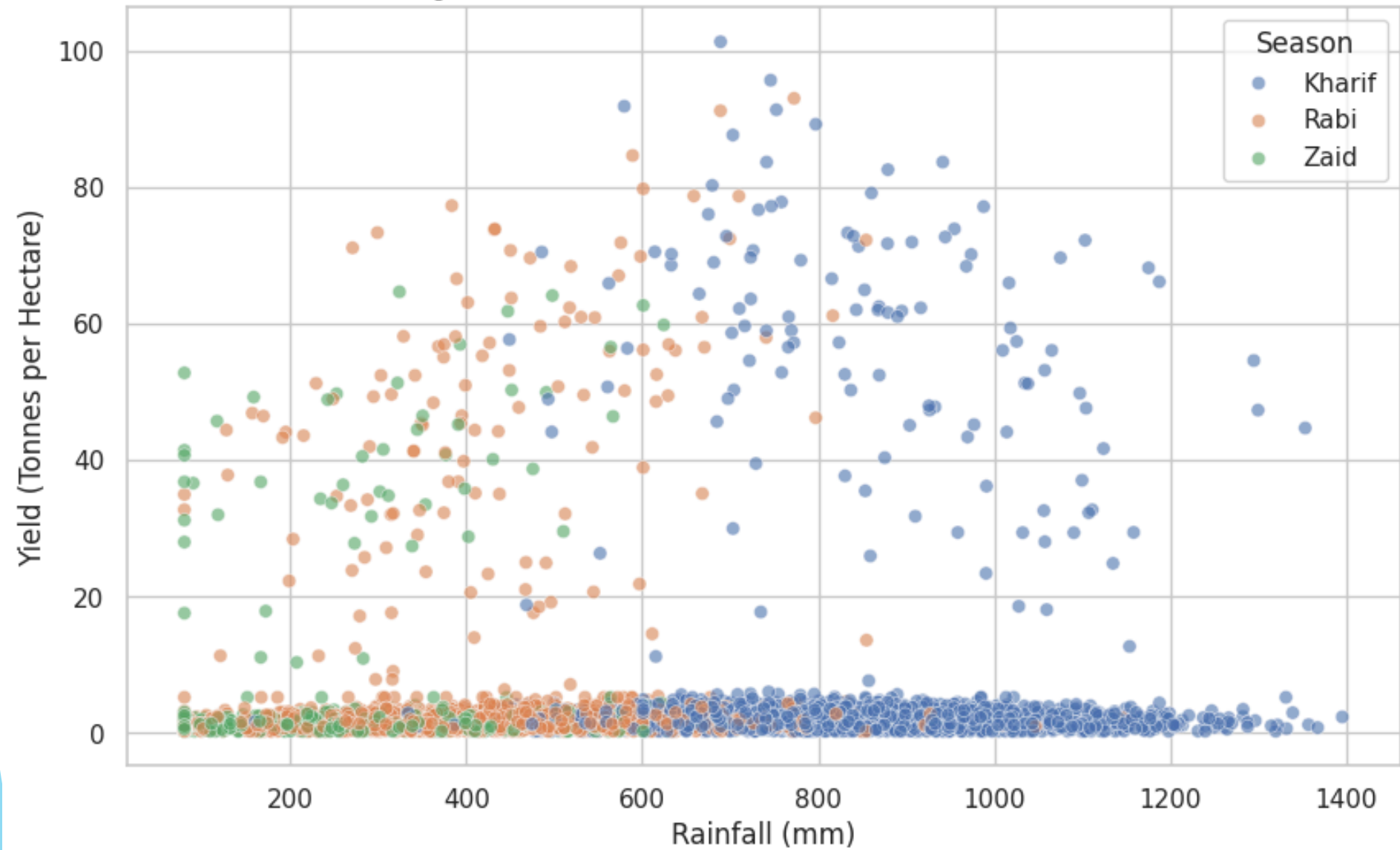


Average Agricultural Profit by Season

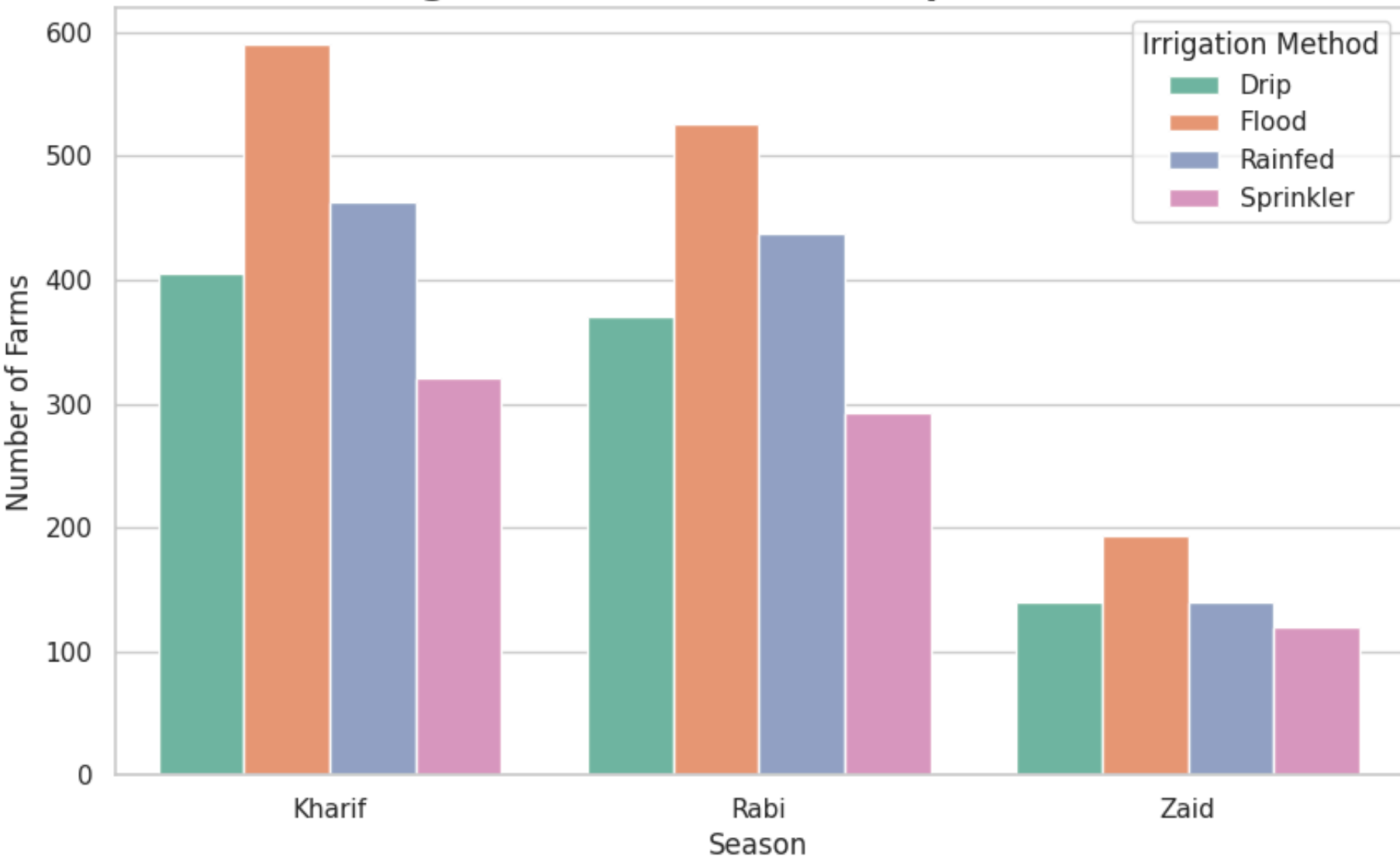


RESULTS

Crop Yield based on Rainfall Variations



Irrigation Methods Chosen per Season



RESULTS

```
print("--- Best and Worst Performing Crops by Season ---")
for season in df['Season'].unique():
    season_data = df[df['Season'] == season]
    crop_profits = season_data.groupby('Crop')['Profit_INR'].mean().sort_values()

    best_crop = crop_profits.index[-1]
    best_profit = crop_profits.iloc[-1]

    worst_crop = crop_profits.index[0]
    worst_profit = crop_profits.iloc[0]

    print(f"\nSeason: {season.upper()}")
    print(f"🏆 Best Crop: {best_crop} (Avg Profit: {best_profit:,.0f} INR)")
    print(f"📉 Worst Crop: {worst_crop} (Avg Profit: {worst_profit:,.0f} INR)")
```

--- Best and Worst Performing Crops by Season ---

Season: KHARIF

🏆 Best Crop: Sugarcane (Avg Profit: 1,000,791 INR)

📉 Worst Crop: Wheat (Avg Profit: -105,872 INR)

Season: RABI

🏆 Best Crop: Sugarcane (Avg Profit: 731,613 INR)

📉 Worst Crop: Wheat (Avg Profit: -119,955 INR)

Season: ZAID

🏆 Best Crop: Sugarcane (Avg Profit: 583,636 INR)

📉 Worst Crop: Rice (Avg Profit: -227,239 INR)

```
print("--- Resource Efficiency: Profit per Cubic Meter of Water ---")
```

```
df['Profit_Per_Water'] = df['Profit_INR'] / df['Water_Used_m3']
```

```
water_efficiency = df.groupby('Season')['Profit_Per_Water'].mean()
```

```
print(water_efficiency.round(2))
```

--- Resource Efficiency: Profit per Cubic Meter of Water ---

Season

Kharif 33.03

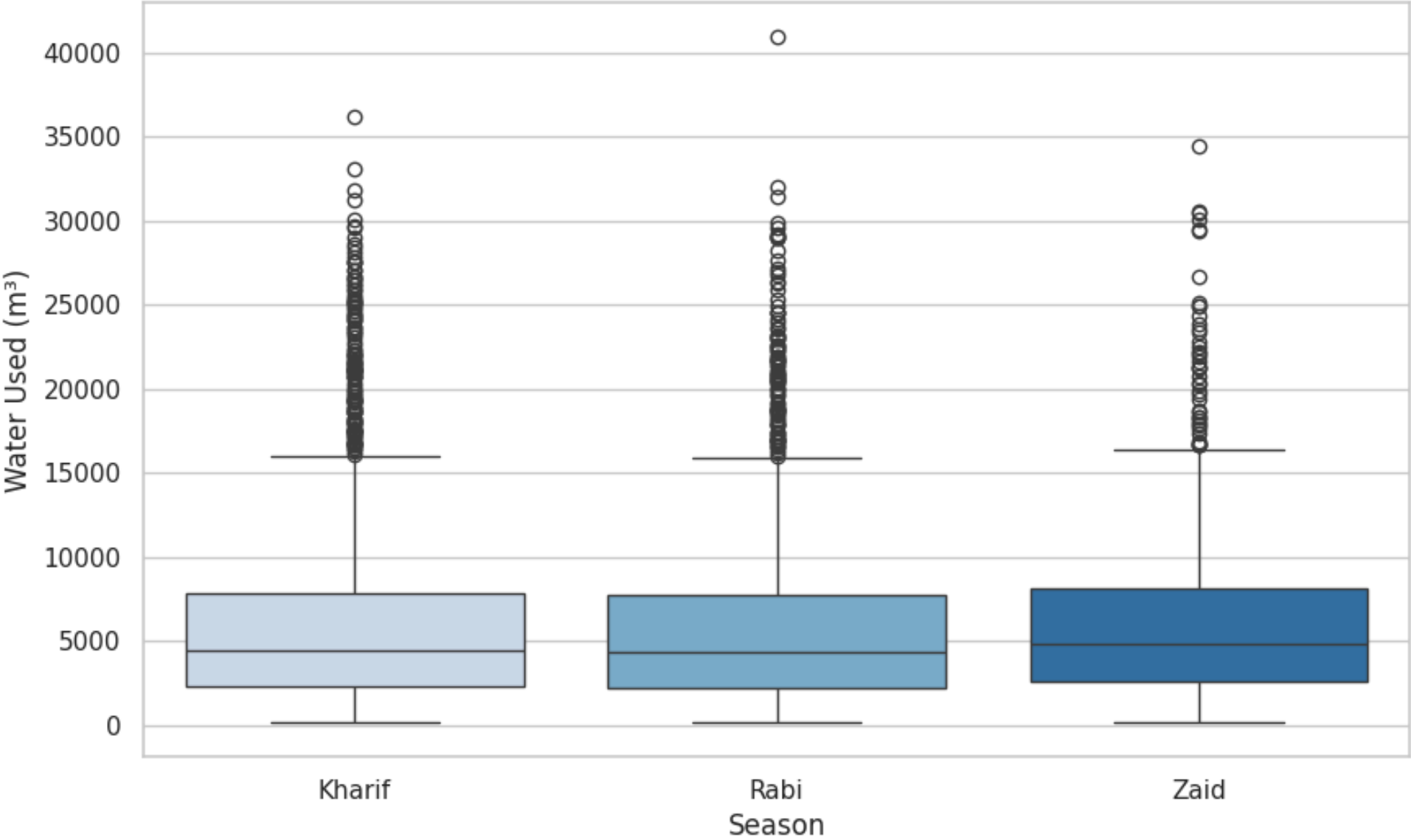
Rabi 12.12

Zaid -12.03

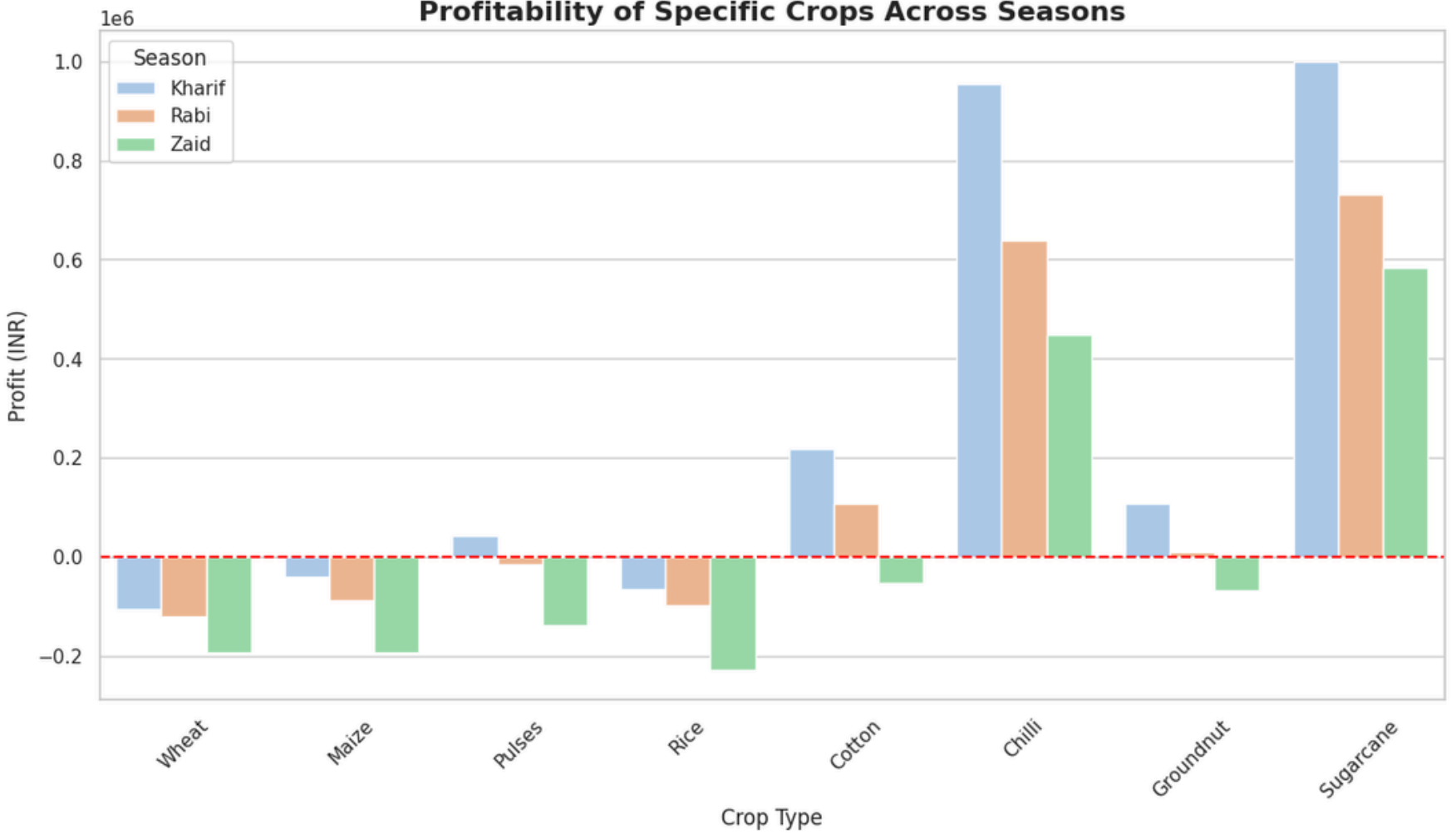
Name: Profit_Per_Water, dtype: float64

RESULTS

Distribution of Water Usage by Season



Profitability of Specific Crops Across Seasons



Future scope



Predictive Machine Learning: Transitioning from this exploratory analysis to building a predictive model that can accurately forecast crop yields and profits based on expected seasonal rainfall and temperature.

Interactive Dashboards: Developing a live, interactive dashboard (using tools like PowerBI or Streamlit) where farmers can filter the data to see customized recommendations for their specific state and crop.

Real-Time Weather Integration: Connecting the analysis to live weather APIs to provide stakeholders with dynamic, real-time alerts about impending seasonal shifts or high pest-risk humidity levels.

Market Demand Analysis: Expanding the dataset to include supply chain and market demand data, allowing for a deeper understanding of why crop selling prices fluctuate between the Kharif, Rabi, and Zaid seasons.



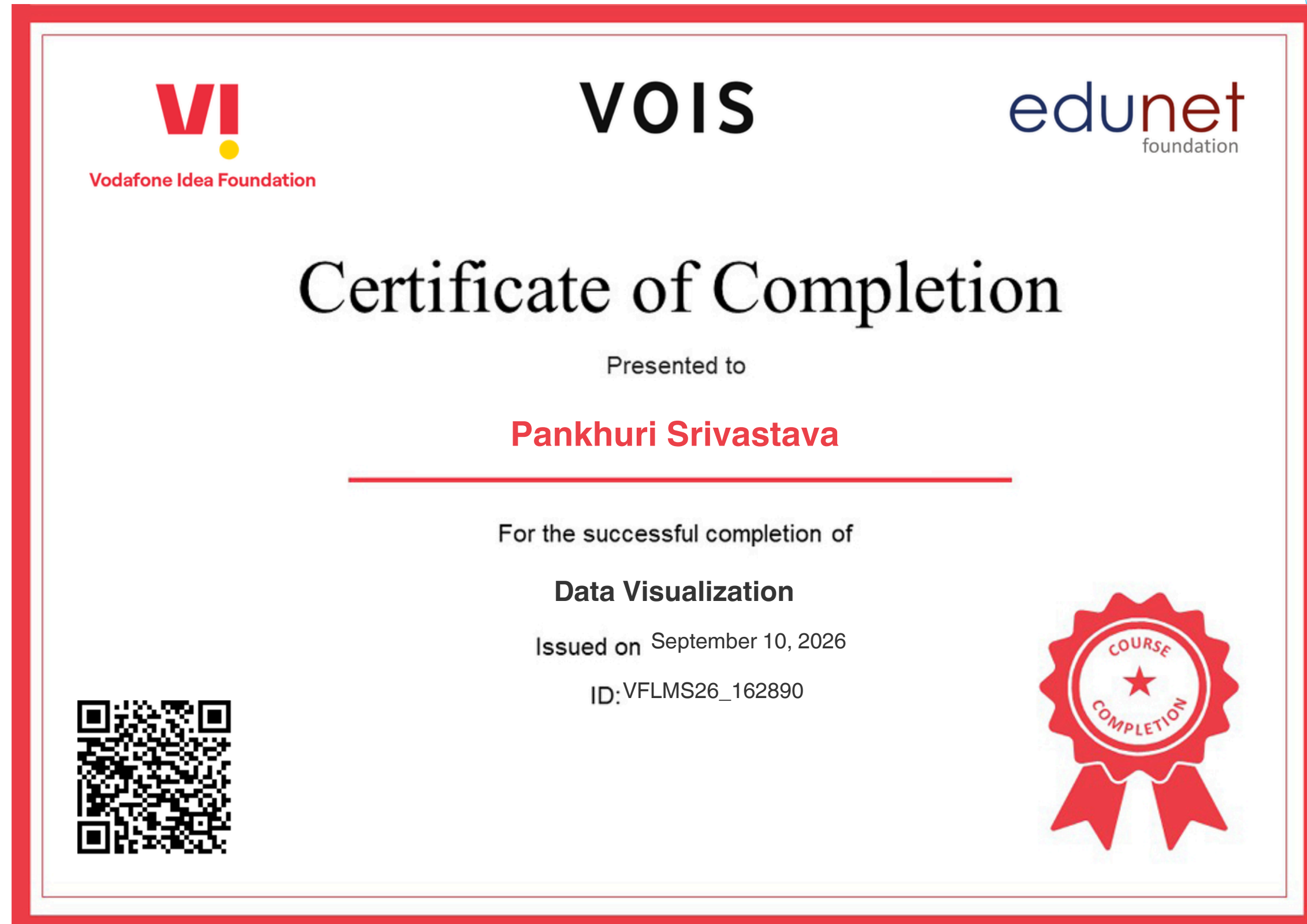
GitHub Link

<https://github.com/SriPankhuri/Seasonal-Agriculture-Performance-Analysis-> 



VOIS Course completion certificate -

Course name- Data Visualization



Thank you



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